Status	Finished
Started	Thursday, 15 October 2026, 4:04 PM
Completed	Thursday, 15 October 2026, 4:12 PM
Duration	8 mins 6 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It divides the list into sublists
- c. It recursively sorts subproblems
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **2**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is difficult to implement
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 40,80,90,150,200,300,400,470,70,120

c. 80,150,200,470,40,90,300,400,70,120

d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming

b. Sorting helps in finding duplicates quickly

c. Sorting makes it harder to search for items

d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **5**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Searching

c. Complexity

d. Merging

The correct answer is:
Complexity

Question **6**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **7**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is less efficient than bubble sort
c. It does not require recursion
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **8**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
b. A sorting algorithm is stable if it preserves the order of duplicate keys
c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. All of the above
- c. Duplicates
- d. Selection

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Heap Sort
- b. Quick Sort
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It avoids the need for recursion
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Insertion Sort
- b. Quick Sort
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Insertion sort
- c. Heap sort
- d. Quick sort

The correct answer is:
Insertion sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Deleting data from a list
c. Searching for data in a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format